

EPILEPSY DIGITAL TWIN REPORT • INTEGRATED NEUROLOGICAL DECISION SUPPORT

# Neuro Digital Twin Integrated Clinical Report

An integrated clinical assessment combining follow-up visits, seizure history, EEG, ECG, CT/MRI imaging, laboratory results, and pharmacogenomic testing for Focal impaired-awareness seizures.

| Patient & Clinical Context                                                                                                                                                                                     |                              | 10/10 data categories included • generated 2026-09-10T16:15:47                                                             |                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| PATIENT<br>John Doe<br>Patient ID SYN-100482                                                                                                                                                                   | AGE / SEX<br>34 years • Male | PRIMARY DIAGNOSIS<br>Focal impaired-awareness seizures<br>Secondary diagnosis: Secondary generalized tonic-clonic seizures | DIAGNOSED SINCE<br>12 May 2021 |
| CURRENT CLINICAL STATUS<br>Stable outpatient epilepsy state with persistent breakthrough seizure risk                                                                                                          |                              |                                                                                                                            |                                |
| CURRENT THERAPY<br>Levetiracetam (Keppra) 1000 mg BID • Lamotrigine 150 mg BID                                                                                                                                 |                              |                                                                                                                            |                                |
| REPORTING PERIOD<br>Data reviewed: 01 Jan – 30 Jun 2026 (~6 months)<br>Clinic visit notes cover 01 Jan 2026 to 02 Mar 2026; laboratory, imaging, EEG/ECG, and questionnaire data continue through 30 Jun 2026. |                              |                                                                                                                            |                                |

|                      |                   |                 |                 |                    |                      |                      |
|----------------------|-------------------|-----------------|-----------------|--------------------|----------------------|----------------------|
| 5<br>Clinical Visits | 12<br>CBC Reports | 6<br>CT Studies | 6<br>MRI Series | 12<br>ECG Sessions | 35<br>EEG Recordings | 12<br>Questionnaires |
|----------------------|-------------------|-----------------|-----------------|--------------------|----------------------|----------------------|

## 01 Executive Summary

Across five clinical-note visits from 2026-01-01 to 2026-03-02, the patient has chronic focal epilepsy with persistent breakthrough seizures on levetiracetam and lamotrigine. The clinician repeatedly documented reasonably stable seizure control and favorable quality-of-life trajectory, while seizure burden, sleep quality, trigger exposure, aura symptoms, and medication tolerability continued to fluctuate.

|                             |                                   |                                      |                                     |
|-----------------------------|-----------------------------------|--------------------------------------|-------------------------------------|
| 67%<br>Overall Health Score | Moderate<br>Overall Clinical Risk | Elevated<br>EEG-Derived Seizure Risk | 10 (7 high)<br>Genetic Safety Flags |
|-----------------------------|-----------------------------------|--------------------------------------|-------------------------------------|

Domain the patient rates worst on questionnaires: Epileptic Triggers. See the Digital Twin Risk Dashboard (Section 13) for the full risk breakdown.

## 02 6-Month Timeline

|             |                                                                                                                                                            |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01 JAN 2026 | Routine epilepsy follow-up with stable clinician impression — Contributes to longitudinal assessment of seizure burden, triggers, sleep, and tolerability. |
| 16 JAN 2026 | Seizure clustering documented during follow-up — Raises near-term seizure-monitoring priority despite otherwise stable clinician impression.               |
| 03 FEB 2026 | CBC/LFT safety review documented — Supports medication-safety monitoring without a documented abnormal lab signal.                                         |
| 15 FEB 2026 | Highest recorded seizure burden in clinical-note window — 11 reported episodes make this the peak burden visit.                                            |
| 02 MAR 2026 | Routine epilepsy follow-up with stable clinician impression — Contributes to longitudinal assessment of seizure burden, triggers, sleep, and tolerability. |
| 10 SEP 2026 | Pharmacogenomic (PGx) panel reported — 76 variants analyzed across 47 medications; see Pharmacogenomics.                                                   |

03 Seizure Trends

Clinical-note episode counts, clustering, and duration

### Reported Episodes per Visit

| Visit Date | Reported Episodes | High-Importance Event |
|------------|-------------------|-----------------------|
| 01 Jan     | 7                 | No                    |
| 16 Jan     | 4                 | Yes                   |
| 03 Feb     | 5                 | No                    |
| 15 Feb     | 11                | Yes                   |
| 02 Mar     | 5                 | No                    |

Red bars mark visits with a high-importance clinical event.

### OVERALL PATTERN

Breakthrough seizure burden fluctuated across visits without a sustained monotonic worsening pattern.

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### HIGHEST BURDEN

11 episodes on 15 Feb 2026

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### LOWEST BURDEN

4 episodes on 16 Jan 2026

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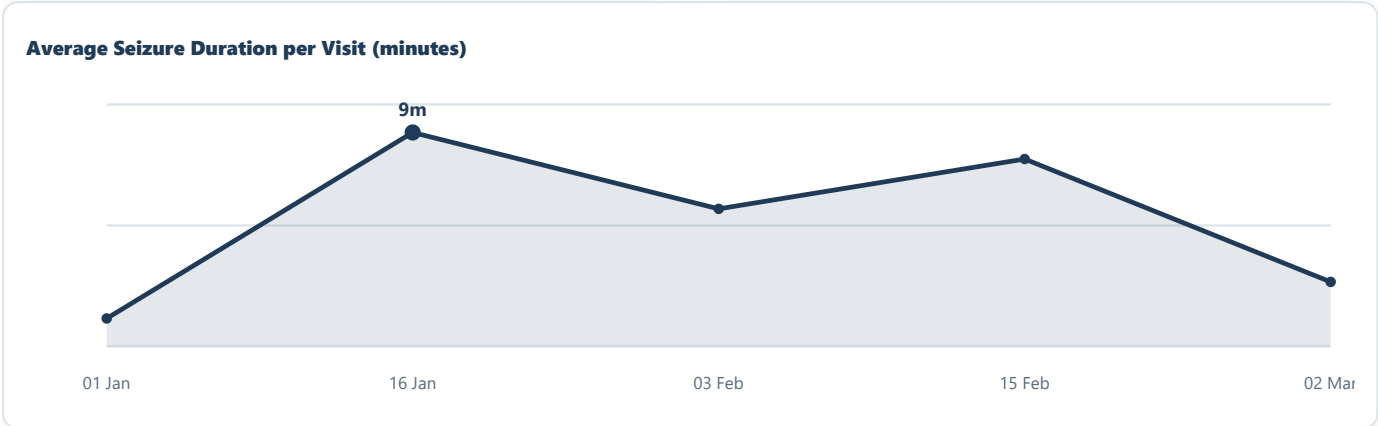
### CLUSTERING

Yes

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### CLINICAL INTERPRETATION

Seizure risk remains clinically relevant because events persist on dual therapy and clustering was documented once.



04 Aura & Triggers

Prodromal patterns and patient-reported trigger exposure

### Persistent Aura / Prodromal Patterns

Deja vu frequency   Epigastric rising sensation   Feeling of derealization/detachment   Metallic taste in mouth   Racing heartbeat prior to episode

Sudden dizziness before seizure onset   Sudden intense anxiety or fear   Tingling or numbness sensations   Unusual smells (olfactory aura)

Visual distortions or flashing lights

Aura and prodromal symptoms remain useful warning signals for episode tracking.

### Aura & Prodromal Symptoms — Questionnaire Domain

Latest score **71.7/100** (first 66.7, trend increasing)

- Did you experience an epigastric "rising" sensation in the stomach: **Frequent**
- Frequency of sudden, unprovoked psychological anxiety or panic feelings (ictal fear): **Rare**
- Frequency of sudden Deja vu or Jamais vu experiences: **Never**
- Did you notice an anomalous metallic, chemical, or bitter taste in your mouth: **Rare sensory distortion**
- Did you experience unprovoked olfactory distortions (strange, non-existent smells): **Never**

### Epileptic Triggers — Lowest-Scoring Domain

Latest score **56.7/100** (min 56.7, trend stable)

- Rate your average psychological stress and workload level this month: **High/Distressing**
- Did you experience a delayed or completely missed dose of your anti-seizure medication: **1 missed dose**
- Exposure to flickering/flashing lights, high-contrast patterns, or prolonged screen glare: **Prolonged exposure**
- Daily average volume of dietary caffeine intake (coffee, tea, energy drinks): **1 cup**
- Any consumption of alcoholic beverages or exposure to hangover states this month: **Occasional drinking**

05 Medication Response

Current regimen, tolerability, and reported side effects

Current Regimen

| DRUG                   | DOSE        |
|------------------------|-------------|
| Levetiracetam (Keppra) | 1000 mg BID |
| Lamotrigine            | 150 mg BID  |

EFFECTIVENESS

Partial response

TOLERABILITY

Generally tolerated with intermittent dizziness and variable side-effect burden

DOCUMENTED TOXICITY

No

CLINICAL INTERPRETATION

Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.

Persistent Side Effects (clinical notes)

Memory fog post-dose

Tremor in hands

Weight change since last visit

Gum hypertrophy

Mood irritability

Nausea after taking medication

Skin rash or itching

Daytime drowsiness

Dizziness/lightheadedness

Unsteady gait (ataxia)

Medication Side Effects — Questionnaire Domain

Latest score **78.3/100** (first 68.3, trend increasing)

- Rate the severity of daytime drowsiness or somnolence: **Mild (Manageable)**
- Experience of dizziness, vertigo, or lightheadedness after taking medication: **Never**
- Presence of physical instability, unsteadiness, or clumsy movements (ataxia): **Slight imbalance**
- Frequency of unprovoked irritability, anger, or mood volatility: **Mild irritability**
- Signs of dermatological distress or unexplained skin rashes: **Clear skin**

06 Pharmacogenomics

76 variants across 47 medications · reported 10 Sep 2026

**PGx summary for the current regimen:** the single most clinically significant finding per drug, with its genetic basis and source citation.

| DRUG          | GENE (GENOTYPE) | WORST / MOST SIGNIFICANT OUTCOME | CITATION      |
|---------------|-----------------|----------------------------------|---------------|
| Lamotrigine   | CYP3A4 (TT)     | REDUCED EFFICACY                 | PMID:28343093 |
| Levetiracetam | SCN1A (CT)      | EFFICACY                         | PMID:28753467 |

External Database Evidence — PubMed / ClinVar / CPIC

Database evidence only, not patient-verified: each row below is a live public-database lookup for the same gene/drug pair as the patient-specific finding in the table above -- it establishes general scientific plausibility for that pair, not a fact confirmed for this individual patient. An UNRESOLVED result means this exact query returned nothing, or failed, at the time checked -- it is not evidence that no relevant data exists, and must not be read as a clean or reassuring negative.

| GENE → DRUG<br>(PATIENT-SPECIFIC PAIR)                                                                              | CPIC                                                                                             | CLINVAR                                                                                                                                                                                                                    | PUBMED                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CYP3A4 → Lamotrigine                                                                                                | UNRESOLVED CPIC lists 'lamotrigine' (RxNorm:28439) but has no CYP3A4-lamotrigine pair on record. | VCV004885026:<br>UNCERTAIN SIGNIFICANCE<br>70 total ClinVar record(s) for this gene (gene-wide count, not variant-specific)<br>ClinVar (NCBI E-utis) · Accession VCV004885026 · Version — · Retrieved 10 Sep 2026 CACHED   | PMID 21635243: Effects of lamotrigine and phenytoin on the pharmacokinetics of atorvastatin in healthy volunteers.<br>PubMed (NCBI E-utis) · PMID 21635243 · Version — · Retrieved 10 Sep 2026 CACHED                                      |
| Note: top table = citation from patient's own PGx report; table below = independent live PubMed search, may differ. |                                                                                                  |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                            |
| SCN1A → Levetiracetam                                                                                               | UNRESOLVED CPIC has no drug entry matching 'levetiracetam'; no gene-drug pair can exist for it.  | VCV004887952:<br>UNCERTAIN SIGNIFICANCE<br>5406 total ClinVar record(s) for this gene (gene-wide count, not variant-specific)<br>ClinVar (NCBI E-utis) · Accession VCV004887952 · Version — · Retrieved 10 Sep 2026 CACHED | PMID 39200163: The Biallelic Inheritance of Two Novel SCN1A Variants Results in Developmental and Epileptic Encephalopathy Responsive to Levetiracetam.<br>PubMed (NCBI E-utis) · PMID 39200163 · Version — · Retrieved 10 Sep 2026 CACHED |
| Note: top table = citation from patient's own PGx report; table below = independent live PubMed search, may differ. |                                                                                                  |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                            |

**Evidence Scope:** live-verified via ClinVar, CPIC, DailyMed, PubMed, and ClinicalTrials.gov only. The rest of the NeuroTwin resource catalog (PharmGKB, DrugBank, Reactome, STRING, GTEx, and others) is out of scope for this live-lookup layer and is not reflected anywhere in this table — see Known Limitations below.

Known Limitations (External Evidence):

- This module queries exactly five public APIs (PubMed, ClinVar, DailyMed, CPIC, ClinicalTrials.gov); it does not cover the rest of the NeuroTwin resource catalog (Reactome, STRING, GTEx, RCSB PDB, etc.) -- those remain out of scope for this module and must not be inferred from it.
- A 'no\_results' status means the API returned zero matching records for the exact query issued; it is not evidence of absence, and must not be reported as a negative finding.
- Responses may be served from the on-disk cache in reports/external\_evidence/.cache/ rather than a fresh call; each record's 'from\_cache' and 'retrieved\_at' fields state which. Pass --refresh to force live calls.

Generated from validated patient data. Decision-support summary only -- not a substitute for clinician review.

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07 Drug Interactions

2 current medication(s) reconciled · Flockhart Cytochrome P450 Drug Interaction Table (curated-snapshot-2024)

Medication Reconciliation

| DRUG                   | DOSE          | STATUS  | FLAGS |
|------------------------|---------------|---------|-------|
| Lamotrigine            | 150.0 mg BID  | current | —     |
| Levetiracetam (Keppra) | 1000.0 mg BID | current | —     |

No conflicting doses found for the same drug across the reconciled source data.

Clinically Relevant Interactions

No clinically required drug-drug interaction was identified for the current regimen.

Essential Monitoring

Continue EEG-derived seizure-risk monitoring.

Continue periodic CBC and safety-laboratory monitoring.

Continue seizure frequency, duration, and clustering tracking.

Review the priority safety flags in the Pharmacogenomics section before adding or changing an interacting medication.

External Database Evidence — DailyMed / ClinicalTrials.gov / PubMed

Database evidence only, not patient-verified: each row below is a live public-database lookup keyed on the drug name from the patient's own current-regimen medication reconciliation above -- it establishes general plausibility for that drug, not a fact confirmed for this individual patient's actual dose or response. An UNRESOLVED result means this exact query returned nothing, or failed, at the time checked -- it is not evidence that no relevant data exists, and must not be read as a clean or reassuring negative.

| DRUG (FROM PATIENT'S CURRENT REGIMEN) | DAILYMED LABEL                                                                                                                                                                                                                                  | CLINICALTRIALS.GOV                                                                                                                                                                                                                                                                           | PUBMED                                                                                                                                                                                                            |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lamotrigine                           | <a href="#">LAMOTRIGINE TABLET, FILM COATED, EXTENDED RELEASE [AMNEAL PHARMACEUTICALS LLC]</a><br>Published Aug 20, 2026<br>DailyMed (NLM REST API v2) · SetID 0833d3df-2df1-4abd-abe0-7fb2c0253c80 · v20 · Retrieved 10 Sep 2026 <b>CACHED</b> | <b>UNRESOLVED</b> No trial matched this exact condition/drug query at the time checked.                                                                                                                                                                                                      | <a href="#">PMID 28862044: Acute lamotrigine overdose: a systematic review of published adult and pediatric cases.</a><br>PubMed (NCBI E-utils) · PMID 28862044 · Version — · Retrieved 10 Sep 2026 <b>CACHED</b> |
| Levetiracetam                         | <a href="#">LEVETIRACETAM TABLET [REMEDYREPACK INC.]</a><br>Published Sep 07, 2026<br>DailyMed (NLM REST API v2) · SetID 2d307850-0ed5-4f32-ac62-0e1e2fe27a7a · v4 · Retrieved 10 Sep 2026 <b>CACHED</b>                                        | <a href="#">NCT07130786: Prophylactic Anti-Seizure Medication vs No Anti-Seizure Medication for Patients With Primary Motor Cortex Brain Metastases</a><br>RECRUITING · 2 total match(es)<br>ClinicalTrials.gov (API v2) · NCT NCT07130786 · Version — · Retrieved 10 Sep 2026 <b>CACHED</b> | <a href="#">PMID 30036676: Levetiracetam.</a><br>PubMed (NCBI E-utils) · PMID 30036676 · Version — · Retrieved 10 Sep 2026 <b>CACHED</b>                                                                          |

Evidence Scope: live-verified via ClinVar, CPIC, DailyMed, PubMed, and ClinicalTrials.gov only. The rest of the NeuroTwin resource catalog (PharmGKB, DrugBank, Reactome, STRING, GTeX, and others) is out of scope for this live-lookup layer and is not reflected anywhere in this table — see Known Limitations below.

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- A 'no\_results' status means the API returned zero matching records for the exact query issued; it is not evidence of absence, and must not be reported as a negative finding.
- Responses may be served from the on-disk cache in reports/external\_evidence/.cache/ rather than a fresh call; each record's 'from\_cache' and 'retrieved\_at' fields state which. Pass --refresh to force live calls.

Limitations of this screen: This assessment is clinical decision support, not an automated prescribing system; no dose, start, or stop decision is made or implied here. No proposed/candidate medication list is present in the source data, so only the current regimen's pairwise and therapy-level evidence is evaluated. Curated pharmacokinetic (CYP) coverage is limited to a small bundled reference table; a drug not listed there is reported as unresolved, not as free of interactions.

35  
Total Recordings

12  
Interictal

11  
Preictal

12  
Ictal

Signal Variance Midpoint by Phase (µV<sup>2</sup>)

344.45

Interictal

4343.59

Preictal

42395.2

Ictal

EEG-derived risk: Elevated

Future seizure probability: Elevated

Dominant feature: Signal Variance

Phase Analysis

| PHASE      | STATE      | ACTIVITY               | VARIABILITY   |
|------------|------------|------------------------|---------------|
| Interictal | Baseline   | Stable                 | Low           |
| Preictal   | Transition | Increasing Instability | Moderate-High |
| Ictal      | Seizure    | Hypersynchronous       | Very High     |

| PHASE      | AMPLITUDE RANGE | STD DEV RANGE | VARIANCE RANGE    |
|------------|-----------------|---------------|-------------------|
| Interictal | -89; 99         | 11.68; 23.5   | 136.44; 552.46    |
| Preictal   | -345; 440       | 14.48; 92.07  | 209.71; 8477.48   |
| Ictal      | -504; 784       | 52.5; 286.42  | 2756.47; 82033.96 |

Longitudinal EEG recordings demonstrate a consistent transition from stable interictal activity to preictal electrical instability followed by high-amplitude ictal seizure activity. Findings are consistent with the dataset's interictal/preictal/ictal phase taxonomy, indicating reproducible seizure-related signal evolution; this is a signal-classification observation, not a clinician-verified diagnosis.

09MRI / CT

Structural imaging: neuroimaging key findings

CT — 6 studies, 12 Jan – 29 Jun 2026

| DATE        | IMPRESSION                                                                                                                              |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 12 Jan 2026 | No acute intracranial abnormality is evident on this single-image model-assisted review.                                                |
| 16 Feb 2026 | No gross acute abnormality is evident on this limited skull-base level image; assessment is limited by slice level and image quality.   |
| 11 Mar 2026 | No acute intracranial abnormality is evident on this single-image model-assisted review.                                                |
| 20 Apr 2026 | No acute intracranial abnormality is evident on this high-convexity image; ventricular assessment is not available at this slice level. |
| 14 May 2026 | No acute intracranial abnormality is evident on this limited-quality near-vertex image.                                                 |
| 29 Jun 2026 | No acute intracranial abnormality is evident. Incidental bilateral choroid plexus calcifications are present.                           |

Across the available CT image files, no gross acute intracranial hemorrhage, midline shift, or mass effect is evident on this model-assisted image-only review. Direct progression assessment is limited because the files represent single slices with different slice levels, resolutions, and image quality. This output is not a radiologist-verified diagnostic report.

MRI — Metadata-supported structural stability, diagnostic interpretation not available

| FINDING                       | STATUS                                                    | DETAIL                                                                                                                                                                                                                                                              |
|-------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Brain Morphology              | NOT DIRECTLY ASSESSED                                     | The provided files include volumetric MRI metadata, but no radiologist interpretation or morphometric segmentation was available for direct morphology assessment.                                                                                                  |
| Brain Volume                  | STABLE WITHIN AVAILABLE METADATA                          | T1w nonzero volume range: 8387642 to 8777268 mm3 (mean 8641065 mm3; range 4.5%). FLAIR nonzero volume range: 6022385 to 6625690 mm3 (mean 6231344 mm3; range 9.7%). These are preprocessing-derived nonzero voxel volumes, not formal brain volumetry.              |
| Tissue Signal Characteristics | QUANTITATIVELY SUMMARIZED, NOT DIAGNOSTICALLY INTERPRETED | FLAIR mean intensity 28.76 to 32.27; standard deviation 47.57 to 54.16; T1w mean intensity 99.82 to 105.56; standard deviation 153.13 to 162.05. Intensity statistics are useful for preprocessing consistency but are not diagnostic tissue-signal interpretation. |

Not directly assessed from the available imaging metadata (this means "not evaluated," not "abnormal"): White Matter, Hippocampus, Cortical Structure, Mass Lesion, Hemorrhage, Infarction.

Stable by available preprocessing metadata. 6 MRI series were available across 3 study identifiers, including 3D T1-weighted MRI, T2-FLAIR MRI. Preprocessed metadata show consistent 3D acquisition characteristics and no metadata-level evidence of major global structural progression. A radiology report, lesion segmentation, hippocampal volumetry, and cortical thickness analysis were not available, so subtle pathology cannot be excluded from these metadata alone.

Generated from validated patient data. Decision-support summary only -- not a substitute for clinician review.

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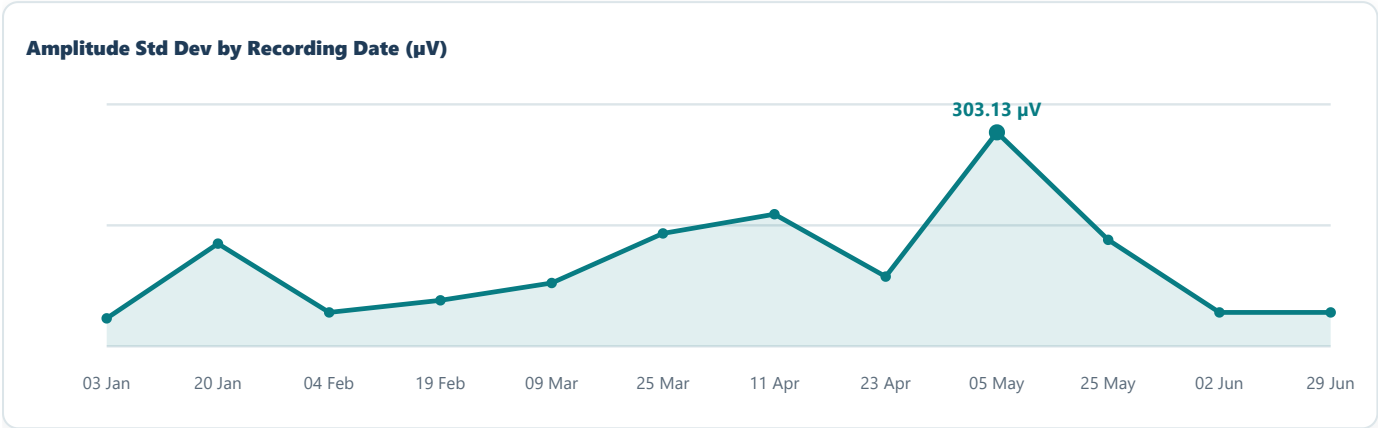
10ECG

single channel ("ECG SD") in every recording

12  
Recordings

99h 11m  
Total Duration

256  
Sampling Rate (Hz)



- All 12 recordings use a single channel sampled at 256 Hz, so statistics are directly comparable across sessions.
  - Recording duration varies considerably, from about 1h 11m (2026-04-11) to about 22h 27m (2026-03-25).
  - The 2026-06-02 and 2026-06-29 recordings have matching percentile values and near-identical histogram shapes, suggesting very similar signal characteristics or a data duplication between the two sessions.
  - Standard deviation of amplitude ranges from about 82.6 uV (2026-01-03) to about 303.1 uV (2026-05-05), indicating substantially different signal variability/noise levels between sessions.
  - Peak-to-peak amplitude (max minus min) ranges from roughly 2,483 uV (2026-05-25) to about 3,577 uV (2026-03-25).
- This report is derived strictly from the statistical fields present in the source dataset. It does not represent a clinical or diagnostic ECG interpretation. Any assessment of cardiac rhythm, heart rate, or abnormality requires review of the underlying waveform by a qualified clinician or an appropriately validated diagnostic algorithm, neither of which is reflected in this report.

11CBC / Labs

12 reports, 01 Jan – 30 Jun 2026

Overall hematological status: Stable within reference ranges. No CBC parameter abnormality was identified against the report-provided reference ranges across the observation period.

| PARAMETER                     | FIRST    | LATEST   | RANGE               | TREND      |
|-------------------------------|----------|----------|---------------------|------------|
| Hemoglobin (Hb)               | 14.4     | 15.1     | 13.5 – 15.9         | increasing |
| Total RBC count               | 5.0      | 5.1      | 4.8 – 5.3           | increasing |
| Packed Cell Volume (PCV)      | 44.7     | 43.5     | 42.2 – 48.4         | decreasing |
| Mean Corpuscular Volume (MCV) | 90.0     | 97.8     | 88.4 – 98.8         | increasing |
| MCH                           | 27.6     | 29.9     | 27.6 – 31.1         | increasing |
| MCHC                          | 33.1     | 33.0     | 32.9 – 34.3         | decreasing |
| RDW                           | 12.1     | 13.5     | 12.1 – 13.5         | increasing |
| Total WBC count               | 5761.0   | 9630.0   | 5476.0 – 9973.0     | increasing |
| Neutrophils                   | 58.0     | 54.0     | 52.0 – 61.0         | decreasing |
| Lymphocytes                   | 22.0     | 29.0     | 22.0 – 34.0         | increasing |
| Eosinophils                   | 4.0      | 4.0      | 1.0 – 5.0           | stable     |
| Monocytes                     | 3.0      | 3.0      | 1.0 – 8.0           | stable     |
| Basophils                     | 1.0      | 1.0      | 0.0 – 2.0           | stable     |
| Platelet Count                | 251725.0 | 209427.0 | 189551.0 – 373102.0 | decreasing |

**Patient-reported vs. measured discrepancy:** the 30 Jun 2026 questionnaire records patient recollections of White Blood Cell (WBC) count baseline trend from your latest Complete Blood Count (CBC): Mild Leukopenia (Borderline low); Platelet Count profile stability from recent blood evaluations: Borderline Thrombocytopenia (Low); Liver Function Test (LFT) - SGOT/AST status: Mild Elevation (40-100 U/L); measured CBC values across all 12 reports in this period remained within their reference ranges throughout (see table below). The discrepancy likely reflects patient recall/perception rather than a measured lab abnormality.

12

Sleep, Cognition & Quality of Life

|                                |                                                |                |                                                                                         |                         |                                                                                                                                         |
|--------------------------------|------------------------------------------------|----------------|-----------------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Sleep                          |                                                | Cognition      |                                                                                         | Quality of Life         |                                                                                                                                         |
| TREND                          | Sleep remained variable across visits          | MEMORY         | Stable on clinician review, with low to intermittent patient-reported memory difficulty | DAILY FUNCTION          | Independence in daily activities was repeatedly normal, intact, or stable                                                               |
| AVERAGE                        | 6.2 hours per night across printed assessments | ATTENTION      | Variable patient-reported concentration and attention symptoms                          | SOCIAL FUNCTION         | Intermittent social withdrawal or isolation symptoms were reported                                                                      |
| Daytime naps needed            |                                                | LANGUAGE       | Mild word-finding difficulty was noted but not described as worsening                   | CLINICAL INTERPRETATION | Quality of life remains affected by seizures, sleep, triggers, and medication tolerability, despite generally stable outpatient status. |
| Difficulty falling back asleep |                                                | MENTAL CLARITY | Improved mental clarity was reported at the latest visit                                |                         |                                                                                                                                         |
| Fatigue level through the day  |                                                |                |                                                                                         |                         |                                                                                                                                         |
| Waking up unrefreshed          |                                                |                |                                                                                         |                         |                                                                                                                                         |

|                                            |             |          |
|--------------------------------------------|-------------|----------|
| Latest Questionnaire Domain Scores (0-100) |             |          |
| Sleep Quality & Architecture               | <div></div> | 91.7/100 |
| Cognitive & Executive Function             | <div></div> | 83.3/100 |
| Lab Biomarkers & Lifestyle                 | <div></div> | 75/100   |

13

Digital Twin Risk Dashboard

Overall clinical risk: Moderate

|                                             |                                            |                                         |                                          |
|---------------------------------------------|--------------------------------------------|-----------------------------------------|------------------------------------------|
| Future Seizure Risk<br>Moderate             | Status Epilepticus Risk<br>Low To Moderate | Hospitalization Risk<br>Low To Moderate | Cognitive Decline Risk<br>Low            |
| Medication Toxicity Risk<br>Low To Moderate | Functional Decline Risk<br>Low             | EEG-Derived Seizure Risk<br>Elevated    | Genetic Safety Flags<br>10 Flags, 7 High |

|                                                                                                                                                                                                                                             |             |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------|
| State Vector (Health Metric Scores, 0-100)                                                                                                                                                                                                  |             |        |
| Disease Stability                                                                                                                                                                                                                           | <div></div> | 67/100 |
| Treatment Response                                                                                                                                                                                                                          | <div></div> | 64/100 |
| Medication Safety                                                                                                                                                                                                                           | <div></div> | 67/100 |
| Cognitive Health                                                                                                                                                                                                                            | <div></div> | 69/100 |
| Sleep Health                                                                                                                                                                                                                                | <div></div> | 66/100 |
| Quality Of Life                                                                                                                                                                                                                             | <div></div> | 67/100 |
| Each domain score is the mean of clinician-documented per-visit ratings across the clinical-note window; overall health score is the mean of the other domains. Derived digital-twin summary metrics, not a direct lab/imaging measurement. |             |        |

14 Clinical Action Plan

Priority Actions (Pharmacogenomics)

- HIGH** **Avoid or use with heightened monitoring**  
See priority\_safety\_flags for the specific gene-drug pairs (ABCB1 (CC), BDNF (TC), CREB1 (AA), FKBP5 (TC), GRIA3 (A), HTR1B (TA), METTL21A (AA)).
- MODERATE** **Dose with caution / consider therapeutic drug monitoring**  
See metabolizer\_profile for reduced-function genes (CYP2C9, CYP2B6, CYP3A5, UGT1A4, POR).
- INFORMATIONAL** **Contextualize**  
Interpret alongside actual clinical history, current medication list, and labs -- this panel alone does not establish a diagnosis.

Recommended Actions

- continue the documented antiseizure medication regimen unless changed by the treating clinician
- maintain the documented plan for consistent sleep schedule and trigger avoidance
- continue seizure and aura tracking across follow-up windows
- continue periodic CBC, liver function, and antiseizure drug-level safety monitoring when scheduled
- reassess at the planned follow-up interval

Monitoring Priorities

- seizure frequency by type
- seizure clustering
- average and longest seizure duration
- post-ictal recovery time
- aura pattern and severity
- sleep duration and sleep fragmentation
- missed medication doses and trigger exposure
- dizziness and other medication side effects
- CBC, liver function, and antiseizure drug-level tracking
- word-finding, memory, and attention symptoms
- Signal Variance
- Amplitude Dynamics
- Electrical Stability
- Preictal Detection
- Seizure Frequency
- Temporal Disease Progression

MRI Follow-Up

- Perform formal radiology review of the MRI image volumes.
- Run automated brain segmentation and longitudinal registration across the three study identifiers.
- Perform hippocampal volumetry because epilepsy-related mesial temporal abnormalities cannot be excluded from metadata alone.
- Run FLAIR white matter lesion analysis if clinically relevant.
- Compare future MRI studies using the same T1w and FLAIR acquisition strategy when possible.
- Correlate MRI findings with seizure semiology and neurological examination.
- Review MRI together with EEG findings for seizure-focus localization.
- Repeat or escalate imaging review if new neurological deficits, worsening seizures, or new focal symptoms occur.

Medication Reconciliation

- Current regimen reconciled and checked against curated reference pharmacology and this patient's own pharmacogenomic findings (Section 07). Predicted-model (SuperCYPsPred) coverage is still unresolved, so a live, comprehensive DDI screen is recommended before adding or changing any interacting medication.